

Worksheet 4: Surfaces & Interiors: Craters, Pressure, & Seismology

Covers Lectures 7 and 8: impact cratering, crater chronology, moment of inertia, central pressure, seismology

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result ("show that..."); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end. All parts are analytical; no computer is needed.

Constants and data

$$G = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

$$\text{Mars: } M = 6.417 \times 10^{23} \text{ kg}, R = 3389.5 \text{ km}, g = 3.71 \text{ m s}^{-2}$$

$$\text{Moon: } R = 1737 \text{ km}, g = 1.62 \text{ m s}^{-2}, \text{ simple-to-complex transition } D_t \approx 15 \text{ km, core radius } 200\text{--}380 \text{ km}$$

$$\text{Mercury: } \bar{\rho} = 5427 \text{ kg m}^{-3}, \text{ core radius fraction } x = r_c/R = 0.83, \text{ core mass fraction } 0.74$$

	Uniform sphere	Moon	Mars	Mercury	Earth
C/MR^2	0.400	0.393	0.364	0.346	0.331

Interior data as in the Lecture 8 notes; crater scaling and chronology as in the Lecture 7 notes. Take $\rho = 3000 \text{ kg m}^{-3}$ for rocky impactors and for the martian target surface.

Problem 1 Impact energetics on Mars

Impact craters are the default landform of the solar system, and a single scaling law connects the size of a crater to the energy of the impactor that made it.

(a) A rocky asteroid of diameter 400 m ($\rho = 3000 \text{ kg m}^{-3}$) strikes Mars at $v = 12 \text{ km s}^{-1}$. Compute its mass and kinetic energy (show that $m \approx 1.005 \times 10^{11} \text{ kg}$ and $E_k \approx 7.238 \times 10^{18} \text{ J}$), then the diameter of the final crater from the gravity-regime scaling

$$D \approx \left(\frac{E_k}{\rho_t g} \right)^{1/4},$$

with ρ_t the target density (show that $D \approx 5.05 \text{ km}$).

(b) Show from the scaling law that, at fixed impact speed and densities, the crater diameter grows with the impactor diameter L as $D \propto L^{3/4}$. Then invert the relation: what impactor diameter does a 50 km martian crater require (show that $L \approx 8.5 \text{ km}$)? Comment on how strongly the fourth root compresses the mapping from impactor size to crater size.

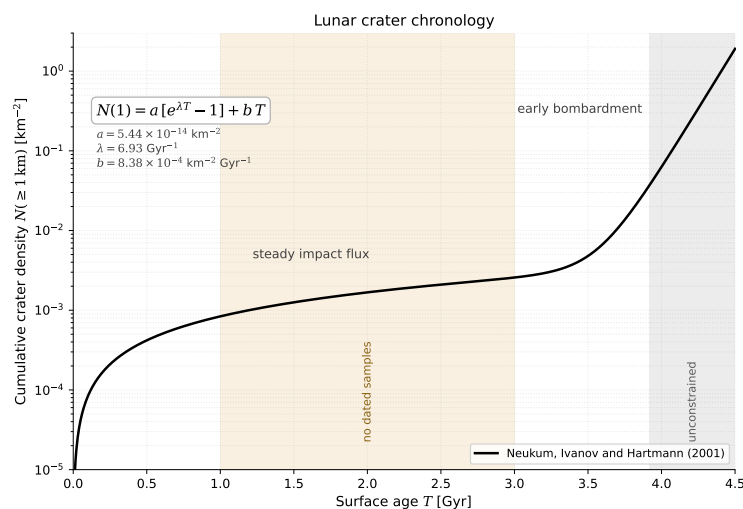
(c) The simple-to-complex transition diameter scales inversely with surface gravity,

$$D_t \approx D_{t,\text{Moon}} \frac{g_{\text{Moon}}}{g}.$$

Locate the transition on Mars (show that $D_t \approx 6.55$ km), classify the crater from part (a) as simple or complex, and explain physically why stronger gravity moves the transition to smaller craters.

Problem 2 Reading ages from craters

Crater counting is the only chronometer that works on every solid surface without a returned sample. The lunar chronology below converts a measured crater density into an absolute age; the coefficients are printed on the figure.



(a) A lunar mare unit has a measured crater density $N(\geq 1 \text{ km}) = 2.5 \times 10^{-3} \text{ km}^{-2}$. On the young branch the linear term dominates, so compute the age from $N \approx bT$ (show that $T \approx 2.983$ Gyr). Then evaluate the exponential term at this age and show that it contributes only about 2% of the measured density, which justifies the approximation.

(b) A heavily cratered highlands unit has $N(\geq 1 \text{ km}) = 3 \times 10^{-2} \text{ km}^{-2}$. Read its approximate age off the figure. Then refine it in two steps: neglect the linear term to obtain a first estimate $T_0 = \lambda^{-1} \ln(1 + N/a)$ (show that $T_0 \approx 3.901$ Gyr), then subtract the linear contribution bT_0 from N and repeat (show that $T_1 \approx 3.885$ Gyr). Why does a factor-of-2 error in the measured N barely change this age, when the same error would double or halve the age in part (a)?

(c) A fellow student writes: “Plains unit A has twice the crater density of plains unit B, so A is twice as old.” Judge this claim: in which regime is it correct, and in which two regimes does it fail?

Problem 3 Moment of inertia and the inside of Mercury

The moment of inertia factor C/MR^2 is the single most informative number about a planet's interior: it measures how strongly mass is concentrated toward the centre, and it is measured from spin precession and gravity without ever drilling a hole.

(a) Model Mercury as a core of uniform density ρ_c inside a mantle of uniform density ρ_m , with core radius fraction $x = r_c/R = 0.83$ and core mass fraction 0.74. From the mean density, compute the two layer densities (show that $\rho_c \approx 7024 \text{ kg m}^{-3}$ and $\rho_m \approx 3295 \text{ kg m}^{-3}$).

(b) The two-layer moment of inertia factor is

$$\frac{C}{MR^2} = \frac{2}{5} \frac{(f-1)x^5 + 1}{(f-1)x^3 + 1}, \quad f = \frac{\rho_c}{\rho_m}.$$

Evaluate it for your Mercury model (show that $C/MR^2 \approx 0.3511$), compare with the measured value 0.346, and state in which direction the two-layer model errs and why.

(c) A fellow student writes: “The Moon’s $C/MR^2 = 0.393$ is within 2% of the uniform-sphere value 0.400, so the Moon cannot have differentiated into a core and a mantle.” Judge this claim, supporting your argument with the two-layer formula and the lunar core radius from the data block.

Problem 4 Central pressure

Hydrostatic equilibrium plus a density profile determines the pressure everywhere inside a planet. The uniform-density estimate is the standard first approximation, and its failure is itself instructive.

(a) Starting from hydrostatic equilibrium,

$$\frac{dP}{dr} = -\frac{G m(r) \rho}{r^2},$$

assume a uniform density ρ and show that

$$P(r) = \frac{2\pi}{3} G \rho^2 (R^2 - r^2), \quad P_c = \frac{2\pi}{3} G \rho^2 R^2 = \frac{3GM^2}{8\pi R^4}.$$

(b) Evaluate the mean density of Mars (show that $\bar{\rho} \approx 3934 \text{ kg m}^{-3}$), then its central pressure (show that $P_c \approx 2.485 \times 10^{10} \text{ Pa}$, about 25 GPa).

(c) For Earth the same estimate gives $P_c \approx 170 \text{ GPa}$, while the true central pressure is about 360 GPa. Explain why the uniform-density formula systematically underestimates the central pressure of a differentiated planet, and state whether your Mars value from part (b) is a lower or an upper bound on the true martian central pressure.

Problem 5 Seismology: listening to interiors

Seismic waves are the sharpest probe of planetary interiors: what arrives, what is missing, and where it is missing all encode the deep structure. No numbers are needed in this problem; the reasoning is the exam skill.

(a) On Earth, direct S waves are absent everywhere beyond about 104° from an earthquake, while P waves are missing only in a ring between about 104° and 140° . Explain the different physical cause of each shadow zone, and state what the S-wave shadow proves about the outer core.

(b) InSight's marsquake data show that at the martian core-mantle boundary the S-wave speed drops to zero while the P-wave speed steps down rather than up. State what each feature implies for the state and the composition of the martian core.

(c) A fellow student writes: "PREM shows the density jumping from about 5570 to about 9900 kg m⁻³ at the core-mantle boundary, but the Adams-Williamson equation predicts a smooth profile there; the equation must be wrong." Judge this claim.